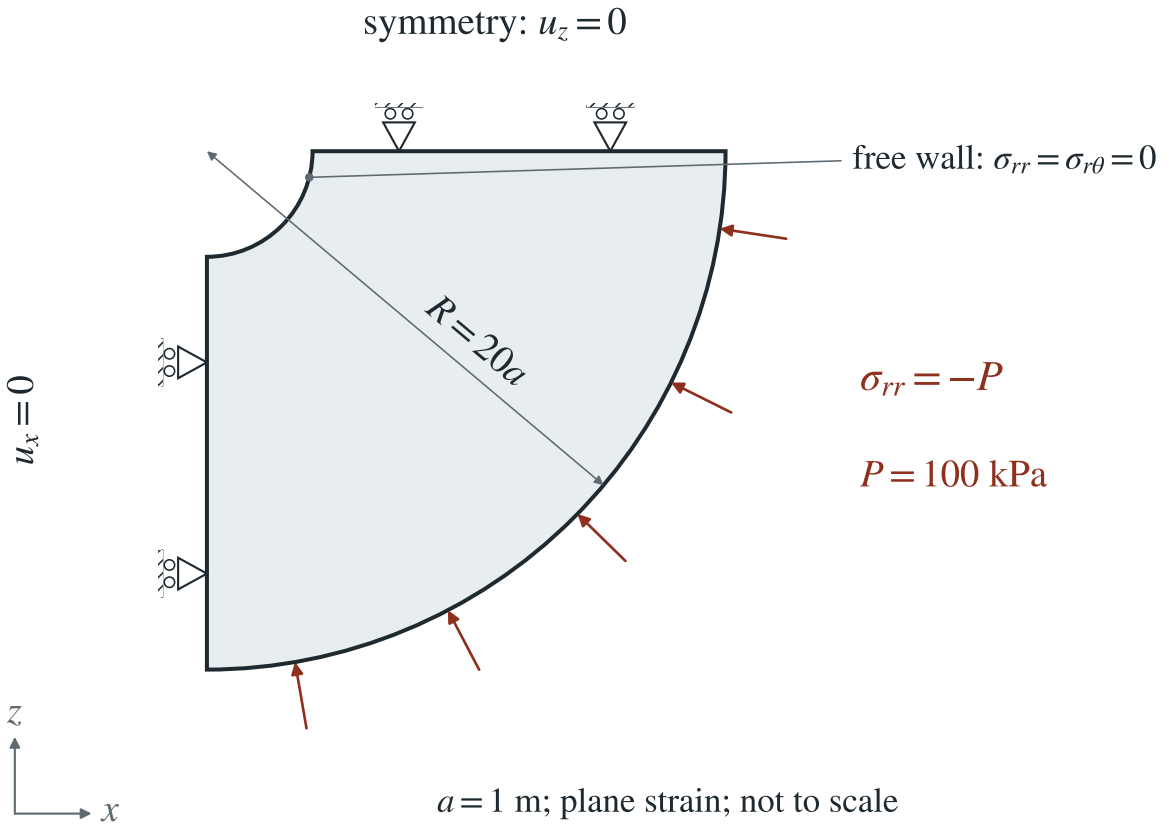
symmetry: $u_z = 0$
free wall: $\sigma_{rr} = \sigma_{r\theta} = 0$
$R = 20a$
$u_x = 0$
$\sigma_{rr} = -P$
$P = 100$ kPa
z
x
$a = 1$ m; plane strain; not to scale